FINAL PROJECT PRESENTATION

Autonomous Driving Perception Stack

Vehicle detection • Multi-object tracking • Lane detection

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A working video-perception prototype, measured and tested.

Driving video contains several perception problems at once

The project combines three complementary outputs into one inspectable pipeline.

01 Detect vehicles

Where are vehicles in each frame?

Output: bounding boxes and class detections.

02 Track identities

Which detections belong to the same object across frames?

Output: track IDs over time.

03 Estimate lanes

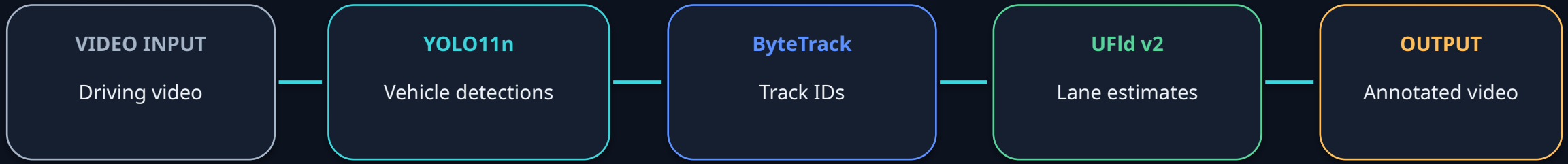
Where are visible lane boundaries?

Output: lane-position overlay.

Engineering objective: integrate all three tasks while keeping CPU processing cost manageable.

A modular pipeline turns video into a combined perception output

Vehicle detection and tracking run on every frame; lane inference is sampled every eight frames.



Detection + tracking: every frame

Lane inference: every 8th frame

Trade-off: reduce lane-model workload while preserving per-frame vehicle detection and tracking.

Three components, one integrated result

Each model or algorithm has a distinct responsibility in the pipeline.

YOLO11n | Detection

- Finds vehicles in each frame
- Produces bounding boxes and detections
- Runs on every input frame

ByteTrack | Tracking

- Associates detections across frames
- Assigns track IDs
- Helps visualize object continuity

Ultra-Fast Lane Detection v2

- Estimates lane positions
- Runs every 8th frame
- Output is combined with detections

Scope boundary: perception and visualization only; no path planning or vehicle control.

The key engineering constraint was CPU inference time

The target was a practical speed improvement without skipping vehicle-detection frames.

Constraint

CPU-only PyTorch environment; model inference dominates runtime.

Experiment

Compared lane inference every 5 frames versus every 8 frames.

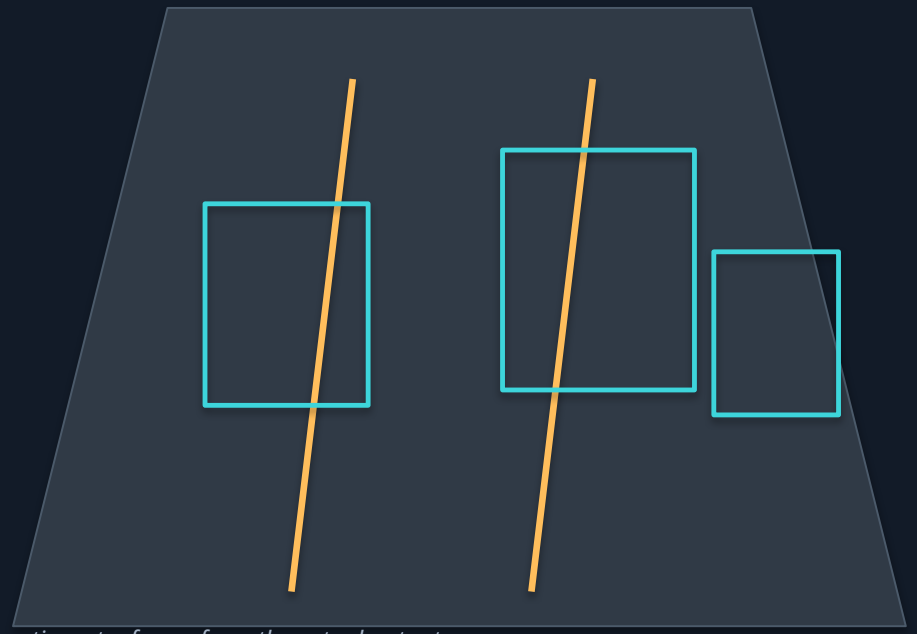
Trade-off

Fewer lane refreshes improve throughput but may reduce responsiveness to road changes.

Remaining visual challenge: lane estimates can degrade on curved road sections.

See the integrated pipeline on real video output

Play the annotated output separately; use this slide to guide the viewer's attention.



Illustrative schematic, not a frame from the actual output.

01 Vehicle boxes

Detected vehicles

02 Track IDs

Identity association across frames

03 Lane overlay

Estimated lane positions

DEMO LINK: INSERT GOOGLE DRIVE LINK TO integrated_traffic.mp4

Measured performance improved after reducing lane-inference frequency

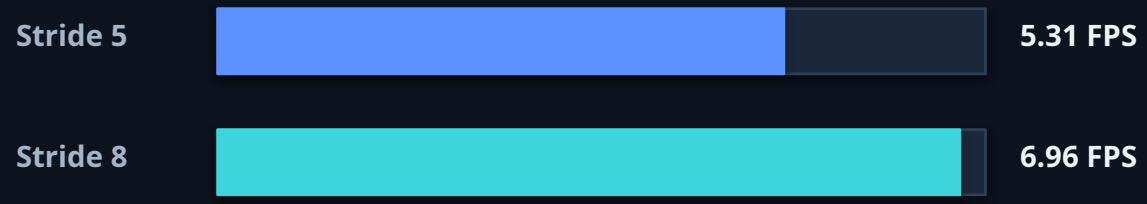
Same input benchmark: 1,260 frames at 1280 × 720 resolution.

6.96
effective FPS

181 s
total processing time

8 / 8
automated tests passed

Effective throughput



+31.1%
throughput improvement
(stride 5 → stride 8)

FPS is measured processing throughput, not the output video's 25 FPS playback rate.

A working prototype is not yet a safety-validated system

Clear limitations make results more credible and guide the next iteration.

Curved roads

Lane estimates can be less accurate on curves than on straight sections.

Compute limits

Current benchmark is CPU-based; real-time performance is not established.

Evaluation scope

Eight tests passed, but broad accuracy and safety evaluation remains outstanding.

Not intended for deployment in a real vehicle or safety-critical decisions.

Prioritize lane quality, measurable evaluation, and acceleration

A practical roadmap for moving from prototype to a more thoroughly evaluated system.

01	**Improve curved-road lanes**	Test alternative lane models and targeted post-processing.
02	**Expand evaluation**	Use diverse videos and ground truth where available.
03	**Optimize inference**	Benchmark GPU acceleration and runtime options.
04	**Measure perception quality**	Report detection, tracking, and lane metrics alongside FPS.

CONCLUSION

One pipeline. Three perception tasks. Measured, tested, and ready to improve.

Integrated

Detection, tracking, and lane estimation in one output.

Benchmarked

1,260 frames; 6.96 effective FPS on the measured setup.

Honest scope

Known limitations documented; next steps identified.

GitHub repository

github.com/SubhamDey2004-coder/autonomous-driving-perception-stack

Thank you